ECON 0100 | Exercise E2 | Competitive Firms

Solutions

Answers and work are shown in the red blocks.

Q1 | Choosing Quantity

The animation's competitive firm has cost functions: $TC = 1000 + q^2$ and $MC = 2q$.

a) If the market price is $P = 80$, find:

- Total revenue from selling 20 units: TR_{20}
- Total revenue selling 21 units: TR_{21}

What is marginal revenue?

b) At $P = 80$, what quantity maximizes the firm's profit? Why not 30 or 50 units?

c) Calculate the firm's maximal profit. (*Hint: ATC at the profit-maximizing quantity is 65.*)

a) $TR_{20} = 1600$, $TR_{21} = 1680$, and marginal revenue is 80. **b)** 40 units. **c)** The maximal profit is 600.

a) $TR = Pq$, so $TR_{20} = 80(20) = 1600$ and $TR_{21} = 80(21) = 1680$. The additional unit adds $MR = TR_{21} - TR_{20} = 80$ to revenue, which is the price. The firm takes the market price as given: selling one more unit does not lower the price received on its other units.

b) Profit is highest where $MC = MR$: $2q = 80$, giving $q = 40$. Not 30 units: there $MC = 60 < MR = 80$, so another unit adds more revenue than cost and increasing output raises profit. Not 50 units: there $MC = 100 > MR = 80$, so the last units cost more than they bring in and decreasing output raises profit. As a check, profit is $80(30) - 1900 = 500$ at 30 units and $80(50) - 3500 = 500$ at 50 units, both below the 600 at 40.

c) Using the supplied $ATC = 65$, profit is $q(P - ATC) = 40(80 - 65) = 600$. Equivalently, $TR = 80(40) = 3200$ and $TC = 1000 + 40^2 = 2600$, so profit is $3200 - 2600 = 600$. It is positive, so operating is better than shutting down.

Q2 | Operating or Leaving

At a price of 40, the best positive output is 20 units. Fixed cost is unavoidable this period but avoidable in future periods by leaving the industry.

a) Calculate profit from operating and from shutting down this period. Which should the firm choose? (*Hint: ATC at the profit-maximizing quantity is 70.*)

b) If this price persists, should the firm stay in the industry in the long run? Explain why this answer can differ from (a).

c) At a price of 40, would firms enter or exit? What happens to market supply and price?

a) Operate: lose 600 instead of 1000. **b)** Exit in the long run if the price persists. **c)** At a price of 40, firms exit, supply decreases, and price rises.

a) At $q = 20$, revenue is $40(20) = 800$, variable cost is $20^2 = 400$, and total cost is $1000 + 400 = 1400$. Operating profit is $800 - 1400 = -600$. With zero output, revenue and variable cost are zero but fixed cost is still 1000, so shutdown profit is -1000. Operating contributes 400 toward that unavoidable cost. Price is above $AVC = 20$, even though it is below $ATC = 70$.

b) A continuing price of 40 does not cover all economic costs. The firm exits when it can avoid future fixed costs. A short-run loss does not by itself imply shutdown; a persistent economic loss does motivate long-run exit.

c) The loss of 600 prompts exit. As firms leave, market supply decreases and price rises, reducing the remaining firms' losses. With identical firms, free entry and exit, and unchanged input prices, exit stops at zero economic profit: $P = MC = \min ATC$. Zero economic profit includes compensation for the owner's opportunity costs.

Q3 | A Higher Marginal Cost

Now suppose every firm's marginal costs go up. Start with identical firms earning zero economic profit in long-run equilibrium. Every unit now costs the same additional amount to produce. Fixed cost and market demand are unchanged, demand slopes down, and entry and exit are free.

a) What happens to price, each firm's output, and profit after the new cost?

b) What happens in the long run to price, each remaining firm's output, total market output, and the number of firms?

The Market



One Firm

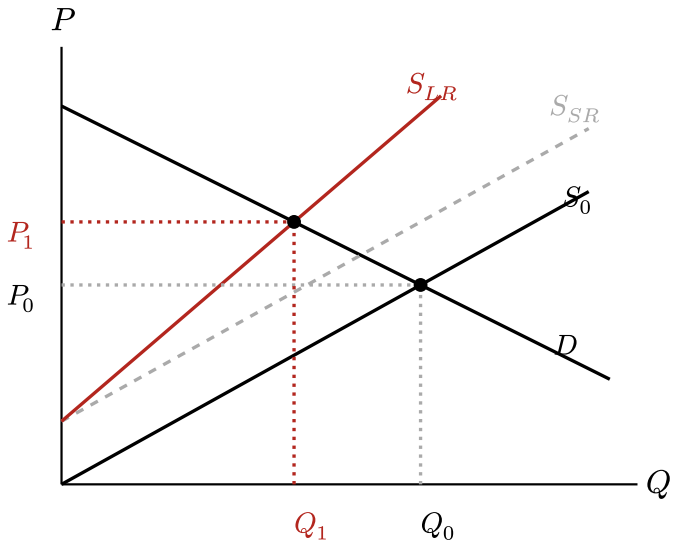


a) Price rises, but by less than the extra cost; each firm's output falls; and profit falls from zero to a loss. **b)** Price is higher, by the full extra cost; each remaining firm's output is back at its original level; total market output is lower; and there are fewer firms.

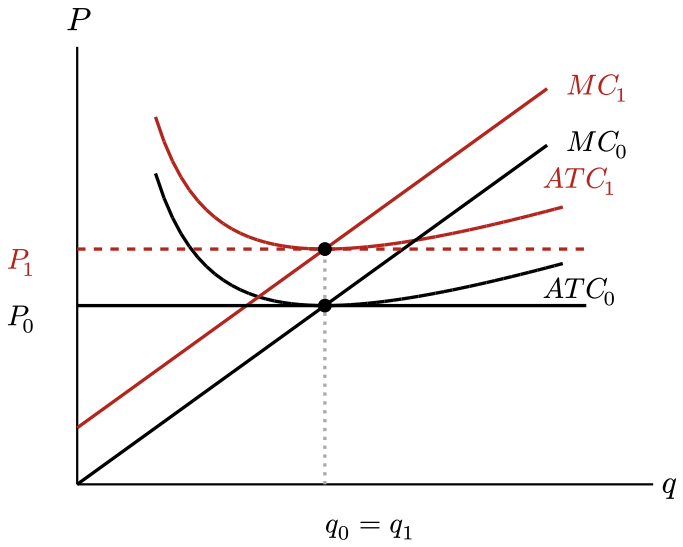
a) Short run, before exit. Write the extra cost per unit as t. Then $TC_1 = 1000 + q^2 + tq$, $MC_1 = 2q + t$, $AVC_1 = q + t$, and $ATC_1 = q + 1000/q + t$. All three per-unit cost curves shift upward in parallel by t. With the same number of firms, market supply also shifts upward by t. The new market price rises by less than t along downward-sloping demand, so each firm reduces its output. Price remains below the new minimum ATC, so firms make losses. They continue operating because price covers AVC at the chosen positive output.

b) Long run, after exit. Losses induce exit, shifting market supply farther inward and raising price. Exit stops at $P_1 = \min ATC_1 = 2\sqrt{1000} + t$. Adding a constant to ATC raises its minimum without changing the quantity at that minimum, so $q_1 = q_0 = \sqrt{1000}$. Survivors expand from their short-run output back to their original scale. Quantity demanded is lower at the higher price, so total market output Q is lower. Since $Q = Nq$ and each remaining firm is back at its original q, the whole fall in total output comes from having fewer firms: $N = Q/q$ is lower.

Market After Exit



Firm Before and After



Black shows the original equilibrium, grey dashed curves show the short-run market supply after the cost increase, and red shows the new long-run equilibrium after exit. The firm's MC and ATC both shift up. Its long-run output returns to the same q_0, at a higher price. The figures use an illustrative uniform per-unit increase; no numerical shock calculation is required.